

Soundness of Ω -logic

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1 Introduction

We recall basic notions and facts related to Ω -logic, mostly based on [BCL06]. In particular, we define $\vdash_\Omega$ and $\models_\Omega$ and present the proof of soundness of Ω -logic.

Proposition 1.1. *Suppose*

1. *there exists a proper class of Woodin cardinals,*
2. *T is a theory,*
3. *ϕ is a sentence.*

If $T \vdash_\Omega \phi$, then $T \models_\Omega \phi$.

Proof. See Proposition 4.6. □

Woodin conjectured that Ω -logic is complete. This is usually called Ω -conjecture.

Conjecture 1.2. *Suppose*

1. *there exists a proper class of Woodin cardinals,*
2. *ϕ is a sentence.*

Then $\vdash_\Omega \phi$ if and only if $\models_\Omega \phi$. □

2 Semantics for Ω -logic

Definition 2.1. Suppose

1. $\mathbb{B}$ is a complete boolean algebra,
2. α is an ordinal.

Then $V_\alpha^\mathbb{B}$ is defined by recursion on α as follows:

- a. if $\alpha = 0$, then $V_\alpha^\mathbb{B} := \emptyset$,
- b. if α is successor, then $V_\alpha^\mathbb{B} := \{f : X \longrightarrow \mathbb{B} : X \subseteq V_{\alpha-1}^\mathbb{B}\}$,
- c. if α is limit, then $V_\alpha^\mathbb{B} := \bigcup_{\xi < \alpha} V_\xi^\mathbb{B}$. □

Proposition 2.2. *Suppose*

1. $\mathbb{B}$ is a complete boolean algebra,
2. α is an ordinal,
3. $\dot{a}$ is a $\mathbb{B}$ -name.

Then¹

$$\sum_{\dot{b} \in V_\alpha^\mathbb{B}} \|\dot{a} = \dot{b}\|_{V^\mathbb{B}} = \|\dot{a} \in R_\alpha\|_{V^\mathbb{B}}.$$

□

Definition 2.3. Suppose

1. $\mathbb{B}$ is a complete boolean algebra,
2. α is an ordinal.

The boolean-valued model $V_\alpha^\mathbb{B}$ is defined by asserting that for all $\dot{a}, \dot{b} \in V_\alpha^\mathbb{B}$,

$$\|\dot{a} \in \dot{b}\|_{V_\alpha^\mathbb{B}} := \|\dot{a} \in \dot{b}\|_{V^\mathbb{B}},$$

$$\|\dot{a} = \dot{b}\|_{V_\alpha^\mathbb{B}} := \|\dot{a} = \dot{b}\|_{V^\mathbb{B}}.$$

□

Lemma 2.4. *Let $\mathbb{B}$ be a complete boolean algebra and let α be an ordinal. Then $V_\alpha^\mathbb{B}$ is a well-defined boolean-valued model.*² □

Proposition 2.5. *Suppose*

1. $\mathbb{B}$ is a complete boolean algebra,

¹Cf. [BCL06, Proposition 1.1].

²Cf. [Jec03, Chapter 14, Sections “Boolean-Valued Models” and “The Boolean-Valued Model V^B ”].

2. α is an ordinal,
3. $\phi(x_0, \dots, x_{n-1})$ is a formula in the language of set theory,
4. $\dot{a}_0, \dots, \dot{a}_{n-1}$ are $\mathbb{B}$ -names.

Then³

$$\|\phi[\dot{a}_0, \dots, \dot{a}_{n-1}]\|_{V_\alpha^\mathbb{B}} = \|R_{\check{\alpha}} \models \phi[\dot{a}_0, \dots, \dot{a}_{n-1}]\|_{V^\mathbb{B}}.$$

□

Notation 2.6. Suppose

1. $\mathbb{B}$ is a complete boolean algebra,
2. α is an ordinal,
3. $\phi(x_0, \dots, x_{n-1})$ is a formula in the language of set theory,
4. $\dot{a}_0, \dots, \dot{a}_{n-1}$ are $\mathbb{B}$ -names.

Then $V_\alpha^\mathbb{B} \models \phi[\dot{a}_0, \dots, \dot{a}_{n-1}]$ iff $\|\phi[\dot{a}_0, \dots, \dot{a}_{n-1}]\|_{V_\alpha^\mathbb{B}} = 1$.

□

Notation 2.7. Suppose

1. $\mathbb{B}$ is a complete boolean algebra,
2. α is an ordinal,
3. T is a theory.

Then $V_\alpha^\mathbb{B} \models T$ iff for all $\phi \in T$, $V_\alpha^\mathbb{B} \models \phi$.

□

Definition 2.8. Suppose

1. T is a theory,
2. ϕ is a sentence.

Then $T \models_\Omega \phi$ iff for all complete boolean algebras $\mathbb{B}$, for all ordinals α , if $V_\alpha^\mathbb{B} \models T$, then $V_\alpha^\mathbb{B} \models \phi$.

□

Lemma 2.9. Suppose

1. T is a theory,
2. ϕ is a sentence.

Then the following are equivalent:⁴

³Cf. [BCL06, Corollary 1.2].

⁴We want to express the intuition that $T \models_\Omega \phi$ is equivalent to

$$\bigwedge T \implies \phi$$

being valid for all models of the form $V_\alpha^\mathbb{B}$.

- a. $T \models_{\Omega} \phi$;
- b. for all complete boolean algebras $\mathbb{B}$ and all ordinals α ,

$$\prod_{\theta \in T} \|\theta\|_{V_{\alpha}^{\mathbb{B}}} \leq \|\phi\|_{V_{\alpha}^{\mathbb{B}}}.$$

□

Theorem 2.10. *Suppose*

- 1. *there exists a proper class of Woodin cardinals,*
- 2. *T is a theory,*
- 3. *ϕ is a sentence,*
- 4. *G is a V -generic filter.*

Then $V \models (T \models_{\Omega} \phi)$ if and only if $V[G] \models (T \models_{\Omega} \phi)$.

Proof. See [BCL06, Theorem 1.8].

□

3 Some Auxiliary Notions and Facts

Definition 3.1. Let A be a universally Baire set and let M be a transitive model. Then M is *A-closed* iff for all posets $\mathbb{P} \in M$,

$$V^{\mathbb{P}} \models (A_g \cap M[g] \in M[g]).$$

□

Lemma 3.2. *Suppose*

1. A is a universally Baire set,
2. κ is an inaccessible cardinal,
3. M is countable and transitive,
4. $\pi : (M, \in, \bar{A}) \longrightarrow (H_\kappa, \in, A)$ is elementary.

Then M is A -closed.

Proof. See [BCL06, Proposition 2.18].

□

Lemma 3.3. *Suppose*

1. A is a universally Baire set,
2. M is an A -closed transitive model of ZFC,
3. $\mathbb{P}$ is a poset in M ,
4. g is V -generic for $\mathbb{P}$.

Then in $V[g]$, $M[g]$ is A_g -closed.

Proof. Let h be a $V[g]$ -generic filter and let us show that $(A_g)_h \cap M[g][h] \in M[g][h]$. Since M is A -closed in V and $g * h$ is a V -generic filter, we have that

$$A_{g*h} \cap M[g * h] \in M[g * h],$$

which is clearly equivalent to what we need.

□

Lemma 3.4 (Woodin). *Suppose there exists a proper class of Woodin cardinals and A is a universally Baire set. Then $(\mathbb{R}, A)^\sharp$ is universally Baire and for every V -generic filter g , $((\mathbb{R}, A)^\sharp)_g = (\mathbb{R}_g, A_g)^\sharp$.⁵*

□

Lemma 3.5. *Suppose*

1. *there exists a proper class of Woodin cardinals,*
2. *δ is a Woodin cardinal,*

⁵This is an unpublished result, but cf. [Lar04, Theorem 3.3.9].

3. G is a V -generic filter for $\mathbb{Q}_\delta$.

Then $\text{UB}^{V[G]}$ is a Wadge-initial segment of $\text{UB}^{\text{Ult}(V,G)}$.

Proof. See [Ste08, Corollary 4.6]. □

Lemma 3.6. *Suppose*

1. *there exists a proper class of Woodin cardinals,*
2. *A is a universally Baire set,*
3. *δ is a Woodin cardinal,*
4. *G is a V -generic filter for $\mathbb{Q}_\delta$.*

Then $j_G^V(A) = A_G$.

Proof. See [Kas25, Lemma 6.6.2]. □

Proposition 3.7. *Suppose*

1. *there exists a proper class of Woodin cardinals,*
2. *A is a universally Baire set,*
3. *$\phi(v_A, v_X)$ is a formula in the language of set theory,*
4. *there exists a poset that forces “ $\exists X \in \text{UB}, L(\mathbb{R}_g, A_g, X) \models \phi[A_g, X]$ ”.*

Then there exists a universally Baire set X so that $L(\mathbb{R}, A, X) \models \phi[A, X]$.⁶

Proof.

- 1° Let g be a V -generic and let $X \in \text{UB}_g$ be such that $L(\mathbb{R}_g, A_g, X) \models \phi[A_g, X]$.
- 2° Let δ be a sufficiently large Woodin cardinal in V and let h be a $V[g]$ -generic filter such that for some V -generic filter G for $\mathbb{Q}_\delta$, $V[g][h] = V[G]$. By Lemma 3.4, $L(\mathbb{R}_{g*h}, A_{g*h}, X_h) \models \phi[A_{g*h}, X_h]$.
- 3° Lemma 3.5 now implies that $\text{Ult}(V, G) \models \exists Y \in \text{UB}, L(\mathbb{R}_G, A_G, Y) \models \phi[A_G, Y]$.
- 4° By Lemma 3.6 and the elementarity of j_G^V , it follows that $V \models \exists Y \in \text{UB}, L(\mathbb{R}, A, Y) \models \phi[A, Y]$. □

⁶Cf. [Ste08, Theorem 5.1].

4 Syntax for Ω -logic

Definition 4.1. Suppose

1. A is a universally Baire set,
2. T is a theory,
3. ϕ is a sentence.

Then $A : T \vdash_{\Omega} \phi$ iff for all A -closed, countable, transitive $M \models \text{ZFC}$ and for all $\alpha < o(M)$, if $R_{\alpha}^M \models T$, then $R_{\alpha}^M \models \phi$. $\square$

Lemma 4.2. Suppose

1. A is a universally Baire set,
2. M is a countable transitive model of ZFC.

Then M is A -closed if and only if for every poset $\mathbb{P} \in M$, there exists a countable collection $\mathcal{D}$ of dense subsets of $\mathbb{P}$ such that every $\mathcal{D}$ -generic filter g for $\mathbb{P}$ is M -generic and satisfies $A \cap M[g] \in M[g]$.

Proof. See [BCL06, Proposition 2.11]. $\square$

Lemma 4.3. Suppose

1. there exists a proper class of Woodin cardinals,
2. A is a universally Baire set,
3. T is a theory,
4. ϕ is a sentence,
5. g is a V -generic filter.

Then $A : T \vdash_{\Omega} \phi$ in V if and only if $A_g : T \vdash_{\Omega} \phi$ in $V[g]$.

Proof. Let $\Phi(v_M, v_A)$ be the formula expressing: “For every poset $\mathbb{P} \in v_M$, there exists a countable collection $\mathcal{D}$ of dense subsets of $\mathbb{P}$ such that every $\mathcal{D}$ -generic filter g for $\mathbb{P}$ is v_M -generic and satisfies $v_A \cap v_M[g] \in v_M[g]$.” Furthermore, let $\Psi(v_A, v_T, v_{\phi})$ be the formula expressing: “For every countable transitive model M of ZFC, if $\Phi(M, v_A)$, then for all $\alpha < o(M)$, if $R_{\alpha}^M \models v_T$, then $R_{\alpha}^M \models v_{\phi}$.” Now, by Lemma 3.4, there exists an elementary embedding

$$j : L(\mathbb{R}, A) \longrightarrow L(\mathbb{R}_g, A_g)$$

such that $j(A) = A_g$, which means that

$$V \models (A : T \vdash_{\Omega} \phi) \iff V \models \Psi[A, T, \phi] \tag{1}$$

$$\iff L(\mathbb{R}, A) \models \Psi[A, T, \phi] \tag{2}$$

$$\iff L(j(\mathbb{R}), j(A)) \models \Psi[j(A), j(T), j(\phi)] \tag{3}$$

$$\iff L(\mathbb{R}_g, A_g) \models \Psi[A_g, T, \phi] \tag{4}$$

$$\iff V[g] \models (A_g : T \vdash_{\Omega} \phi). \tag{5}$$

$\square$

Definition 4.4. Let T be a theory and let ϕ be a sentence. Then $T \vdash_{\Omega} \phi$ iff there exists a universally Baire set A such that $A : T \vdash_{\Omega} \phi$. $\square$

Proposition 4.5. *Suppose*

1. *there exists a proper class of Woodin cardinals,*
2. *T is a theory,*
3. *ϕ is a sentence,*
4. *g is a V -generic filter.*

Then $T \vdash_{\Omega} \phi$ in V if and only if $T \vdash_{\Omega} \phi$ in $V[g]$.

Proof.

- 1° The implication $(\Rightarrow)$ is immediate from Lemma 4.3. Let us verify $(\Leftarrow)$.
- 2° We are working in $V[g]$ and assuming that $T \vdash_{\Omega} \phi$.
- 3° Let $\Psi(v_A, v_T, v_{\phi})$ be as in the proof of Lemma 4.3. Since $V[g] \models \exists A \in \text{UB}, A : T \vdash_{\Omega} \phi$, it follows that $V[g] \models \exists A \in \text{UB}, L(\mathbb{R}_g, A) \models \Psi[A, T, \phi]$.
- 4° By Proposition 3.7, it holds in V that for some universally Baire set A , $L(\mathbb{R}, A) \models \Psi[A, T, \phi]$. Hence, $V \models (T \vdash_{\Omega} \phi)$.

$\square$

Proposition 4.6. *Suppose*

1. *there exists a proper class of Woodin cardinals,*
2. *T is a theory,*
3. *ϕ is a sentence,*
4. *$T \vdash_{\Omega} \phi$.*

Then $T \models_{\Omega} \phi$.

Proof.

- 1° Let $\mathbb{B}$ be a complete boolean algebra and let α be an ordinal so that $V_{\alpha}^{\mathbb{B}} \models T$. We want to see that $V_{\alpha}^{\mathbb{B}} \models \phi$.
- 2° Let A be a universally Baire set such that $A : T \vdash_{\Omega} \phi$, let λ be an inaccessible cardinal such that $\mathbb{B}, \alpha \in V_{\lambda}$, and let

$$\pi : (M, \in, T, \bar{\mathbb{B}}, \bar{\alpha}, \bar{A}) \longrightarrow (H_{\lambda}, \in, T, \mathbb{B}, \alpha, A)$$

be elementary, with M countable and transitive. By Lemma 3.2, M is A -closed.

3° **Claim.** $M_{\bar{\alpha}}^{\bar{\mathbb{B}}} \models \phi$

Proof.

1' Let us assume otherwise. Then there exists $b \in \bar{\mathbb{B}}$ such that

$$b \Vdash_{\bar{\mathbb{B}}}^M R_{\bar{\alpha}} \models \neg\phi.$$

2' Since $V_{\alpha}^{\mathbb{B}} \models T$, it follows by elementarity that $\Vdash_{\bar{\mathbb{B}}}^M R_{\bar{\alpha}} \models T$.

3' Now, let g be V -generic for $\bar{\mathbb{B}}$ that contains b and let us work in $V[g]$.

Observe that $R_{\bar{\alpha}}^{M[g]} \models T + \neg\phi$.

4' Lemma 4.3 implies that $A_g : T \vdash_{\Omega} \phi$.

5' Lemma 3.3 implies that $M[g]$ is A_g -closed.

6' Thus, if $R_{\bar{\alpha}}^{M[g]} \models T$, then $R_{\bar{\alpha}}^{M[g]} \models \phi$. This is a contradiction.

□

4° By elementarity, $V_{\alpha}^{\mathbb{B}} \models \phi$, as required.

□

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